

CS-AML

Product & Feature Specification

Version 0.1

Civil Society Financial Intelligence / AML Investigation Platform

Status: Normative product baseline / implementation specification

Product axiom

The platform SHALL help analysts discover, structure, test, explain, review, and disseminate financial intelligence without converting uncertainty into fact or replacing accountable human judgement.

Document Control

Field	Value
Document	CS-AML Product & Feature Specification
Version	0.1
Status	Normative product baseline
Primary audience	Product owners, investigators, analysts, architects, developers, security/privacy reviewers, QA
Normative inputs	CS-AML Framework; Goals & Non-Goals; Typology Catalogue; Investigation Methodology; Data Model Specification; Control Implementation Guide; Technology Architecture
Normative language	SHALL/MUST = mandatory; SHOULD = recommended; MAY = optional
Primary output	Prioritised, testable product backlog and release scope

1. Purpose and Product Definition

This specification translates the CS-AML normative framework into a buildable software product. It defines the product boundary, user roles, capabilities, functional features, acceptance criteria, integrations, non-functional requirements, release priorities, and traceability expectations for a Civil Society Financial Intelligence / AML Investigation Platform.

Product definition

CS-AML Platform is a civil-society financial intelligence and investigation system. It is NOT a bank transaction-monitoring system, a law-enforcement case system, a guilt-determination engine, or a general-purpose mass-surveillance platform.

The product SHALL enable a structured chain:

SOURCE → EVIDENCE → CLAIM/FACT → INDICATOR → HYPOTHESIS → ASSESSMENT → INTELLIGENCE PRODUCT

The product SHALL preserve reusable truth-bearing objects—especially Evidence, Entity, Relationship, Asset, Event, and ValueFlow—across cases while preserving case compartmentalisation and access controls.

2. Product Goals and Success Outcomes

ID	Goal	Outcome
PG-01	Structure investigations	Replace ad-hoc folders, spreadsheets, screenshots, and disconnected notes with a controlled investigation workspace.
PG-02	Preserve provenance	Make every material analytical conclusion traceable to source and evidence.

ID	Goal	Outcome
PG-03	Follow the value	Support direct, documented, reconstructed, and hypothetical value-flow analysis without pretending public-source inference is bank-transaction evidence.
PG-04	Resolve entities safely	Allow identity/entity consolidation while retaining confidence, evidence, reversibility, and audit history.
PG-05	Test hypotheses	Support competing explanations, disconfirming evidence, unknowns, and confidence-based assessments.
PG-06	Apply typologies carefully	Use typology catalogues as analytical references, not proof of wrongdoing.
PG-07	Create decision-useful intelligence	Generate reviewable intelligence notes, network analysis, referral packages, and publication-ready outputs.
PG-08	Protect people and sources	Apply minimisation, compartmentalisation, least privilege, source protection, secure dissemination, and retention controls.
PG-09	Enable collaboration	Support accountable multi-analyst workflows, peer review, structured handover, and controlled partner sharing.
PG-10	Remain extensible	Integrate specialised external tools rather than rebuilding mature capabilities unnecessarily.

3. Product Non-Goals

ID	Non-goal	Rule
NG-01	No autonomous guilt determination	The system SHALL NOT determine that a person committed money laundering or another crime.
NG-02	No unrestricted surveillance	The product SHALL NOT be designed for bulk collection of people unrelated to a defined investigation purpose.
NG-03	No unauthorised acquisition	The product SHALL NOT provide hacking, credential theft, unauthorised interception, or covert access functions.
NG-04	No bank-core replacement	The product is not a bank AML transaction-monitoring or customer-onboarding system.
NG-05	No evidence laundering	AI, graph scores, data enrichment, or imported intelligence SHALL NOT silently become verified fact.
NG-06	No forced monolith	Search, graph, OCR, sanctions screening, and blockchain analytics MAY be delegated to external components.

4. User Roles and Personas

Role	Primary responsibility
Investigator / Analyst	Opens and develops cases; registers sources/evidence; creates entities, relationships, events, flows, hypotheses, and assessments.
Case Owner / Investigation Lead	Owns scope, lifecycle gates, prioritisation, escalation, closure, and accountability.
Reviewer / Red-Team Analyst	Independently tests identity resolution, evidence basis, alternative explanations, confidence, and wording.
Evidence Custodian	Preserves originals, hashes, derivative lineage, retention status, and evidence integrity.
Data Steward	Maintains entity quality, controlled vocabularies, merge/split history, and data-quality remediation.
Privacy / Legal Reviewer	Reviews sensitive collection, personal-data necessity, publication/referral risk, retention, and lawful access.
Security Administrator	Manages IAM, access policy, secrets, audit, secure storage, backup, incident response, and system hardening.
Partner / External Reviewer	Receives only authorised cases/products/objects under explicit sharing controls.
Platform Administrator	Manages configuration, connectors, taxonomy, templates, feature flags, and system operations without automatic access to case content.
Auditor / Assurance Reviewer	Tests controls and reconstructs material analytical decisions without altering investigation content.

5. Product Design Principles

Principle	Product requirement
Evidence before allegation	The UX SHALL visually distinguish source statements, verified facts, analyst inference, typology indicators, hypotheses, and assessments.
Case is context	Case controls scope, permissions, questions, tasks, review, and products; reusable entities/evidence SHALL not be duplicated merely because they appear in multiple cases.
Human accountability	Material merges, adverse assessments, dissemination, and publication SHALL require attributable human actions.
Uncertainty is data	Confidence, unknowns, dispute, ambiguity, and alternative explanations SHALL be representable as first-class information.
Derived is not canonical	Search indexes, graph projections, embeddings, AI summaries, and caches SHALL be rebuildable from canonical records.
Security by default	Least privilege, sensitive-source compartmentalisation, auditability, secure export, and retention SHALL be default product behaviours.
Build the unique layer	CS-AML SHALL prioritise its unique investigation methodology over reimplementing commodity OCR, graph database, sanctions datasets, or blockchain ingestion.

6. End-to-End Product Workflow

1. Lead / tip intake
2. Triage and legitimacy check
3. Case charter and investigation question
4. Collection plan
5. Lawful source collection
6. Evidence registration and preservation
7. Document processing / extraction
8. Entity resolution
9. Relationship / ownership / asset mapping
10. Timeline construction
11. Value-flow reconstruction
12. Typology and indicator mapping
13. Hypothesis testing
14. Assessment and confidence
15. Peer / legal / security review
16. Intelligence product
17. Referral / controlled sharing / publication
18. Closure / monitoring / retention

7. Product Capability Map

ID	Capability	Scope
CAP-01	Case & Workflow Management	Case, charter, scope, tasks, lifecycle gates, status, assignment, review
CAP-02	Source & Evidence Management	Source register, evidence store, hashes, extracts, derivatives, provenance
CAP-03	Document Intelligence	Upload, OCR, parsing, metadata, extraction candidates, deduplication
CAP-04	Entity & Identity Management	Registry, aliases, identifiers, candidate matching, merge/split, confidence
CAP-05	Relationship / Ownership / Asset	First-class relationships, beneficial ownership, control, assets, attribution
CAP-06	Timeline & Events	Events, temporal precision, chronology, visual timeline
CAP-07	Value-Flow Analysis	Direct/documented/reconstructed/hypothetical flows, multi-leg flows, flow graph
CAP-08	Typology & Indicator Analysis	Typology catalogue, indicators, counter-indicators, mapping, consistency level
CAP-09	Hypothesis & Assessment	Competing hypotheses, support/contradiction, gaps, confidence, judgement
CAP-10	Search / Graph / Analytics	Full-text, faceted search, graph navigation, paths, clusters, network analytics
CAP-11	Intelligence Products & Review	Reports, peer review, red-team review, approval, correction/versioning
CAP-12	Dissemination & Referral	Export, handling labels, recipient controls, referral packages, sharing log

ID	Capability	Scope
CAP-13	Administration & Governance	Roles, vocabularies, templates, policies, retention, feature flags
CAP-14	Security / Audit / Operations	IAM, MFA, audit, backup, monitoring, incident response, DR
CAP-15	Integrations & Automation	OpenAleph, OpenSanctions, Flowintel, GraphSense, graph/search, AI assist

Priority model

P0 = MVP mandatory; P1 = Phase 2 / operational depth; P2 = Phase 3 / advanced intelligence. A feature can be technically sophisticated yet remain P2 if the core methodology works without it.

8. Functional Feature Catalogue

The following catalogue is the product baseline. IDs are stable within v0.1 and SHOULD be used in issue trackers, architecture decisions, test cases, and release notes.

CAP-01 — Case & Workflow Management

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-CASE-001	Case creation and register	P0	Investigator	Create a case with stable ID, title, owner, purpose, classification and status.	Case	GOV-01,CAS-01	Case cannot become Active without owner, purpose and investigation question.
F-CASE-002	Investigation Charter	P0	Case Owner	Define question, scope, exclusions, jurisdictions, period, risks, authorised/prohibited collection and intended outputs.	Case, InvestigationQuestion	CAS-01,PRI-01	Charter version is preserved; scope changes are recorded and attributable.
F-CASE-003	Lifecycle gates G0-G6	P0	Case Owner	Advance a case only when required review gates are satisfied.	Case, Review, AuditEvent	CAS-02,QUA-01	High-risk gate cannot self-approve where policy requires independent review.
F-CASE-004	Tasks and assignments	P0	Investigator	Assign collection, verification, analysis and review tasks with due dates and status.	Task, Case	GOV-01	Task ownership, completion and history are visible.
F-CASE-005	Case activity timeline	P0	All roles	See significant case actions in chronological order.	AuditEvent, Case	AUD-01	Material actions appear with actor and timestamp.
F-CASE-006	Case-to-case links	P1	Analyst	Link related cases without exposing restricted contents automatically.	CaseRelationship	SEC-01	Cross-case link respects access policy and does not grant implicit access.

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-CASE-007	Monitoring / reopen state	P1	Case Owner	Close, monitor, or reopen cases with rationale.	Case	CAS-01,AUD-01	Closure reason and reopen event are immutable history.

CAP-02 — Source & Evidence Management

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-EVD-001	Source register	P0	Investigator	Register source origin, publisher, URL/location, dates, access method, reliability and legal/access note.	Source	SRC-01,SRC-02	Material source cannot support assessment without provenance fields.
F-EVD-002	Evidence upload and original preservation	P0	Evidence Custodian	Store original evidence separately from working copies.	EvidenceItem	EVD-01,SEC-01	Original is read-only to analysts; replacement creates a new version.
F-EVD-003	Hash and integrity record	P0	Evidence Custodian	Compute and verify cryptographic hash for critical files.	EvidenceItem	EVD-01	Stored hash can be recomputed and compared.
F-EVD-004	Evidence extract / citation	P0	Investigator	Create page/paragraph/region extracts linked to original evidence.	EvidenceExtract	EVD-02	Each extract links to parent evidence and precise location.
F-EVD-005	Derivative lineage	P0	Investigator	Track OCR, translation, cropped image, parsed table or transformed dataset as derivative.	EvidenceItem	EVD-02	Derivative records parent, transformation , creator/tool and time.
F-EVD-006	Source reliability & information credibility	P0	Analyst	Rate source reliability separately from information credibility.	Source,Claim	SRC-02,ASM-01	UI prevents one rating from silently standing for both dimensions.
F-EVD-007	Evidence legal hold / retention state	P1	Evidence Custodian	Apply retention, legal hold and disposition state.	EvidenceItem	PRI-02	Deletion workflow respects hold and produces audit record.

CAP-03 — Document Intelligence

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-DOC-001	Document ingestion	P0	Investigator	Upload PDF, DOCX, XLSX, CSV, image and text evidence.	EvidenceItem	EVD-01	Unsupported/ failed ingest does not alter original.

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-DOC-002	OCR / text extraction	P1	Investigator	Extract searchable text while preserving original and page mapping.	EvidenceItem, DerivedArtifact	EVD-02,TEC-01	OCR output is marked derived and page-linked.
F-DOC-003	Structured table extraction	P1	Analyst	Extract tabular data as reviewable derivative records.	DerivedArtifact	EVD-02,TEC-01	Analyst can compare extracted cells to original.
F-DOC-004	Entity candidate extraction	P1	Analyst	Generate person/company/address/identifier candidates from documents.	EntityCandidate	TEC-01	Candidates never become canonical entities without human confirmation or configured rule.
F-DOC-005	Duplicate document detection	P1	Evidence Custodian	Detect exact/near duplicate files and preserve provenance for each source occurrence.	EvidenceItem	EVD-01	Deduplication never drops separate provenance.
F-DOC-006	Translation assist	P2	Analyst	Create linked translations with source text retained.	DerivedArtifact	EVD-02,TEC-01	Translation is marked derived and reviewable.

CAP-04 — Entity & Identity Management

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-ENT-001	Entity registry	P0	Analyst	Create canonical person, organisation, account, wallet, address, domain and other entity records.	Entity	ENT-01	Each entity has type, canonical name and provenance-linked assertions.
F-ENT-002	Aliases and identifiers	P0	Analyst	Record aliases, registration numbers, IDs, phones, emails and external IDs with source.	Entity,Identifier	ENT-01	Identifier provenance is visible and conflicting values can coexist.
F-ENT-003	Candidate matching	P0	Analyst	Compare possible duplicate entities using discriminating features.	EntityMatchCandidate	ENT-01	System shows both matching and conflicting attributes.
F-ENT-004	Merge / unmerge	P0	Data Steward	Merge with rationale and later reverse while preserving history.	Entity,MergeDecision	ENT-01,AUD-01	Unmerge restores prior identities and relationships without history loss.

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-ENT-005	Entity confidence/status	P0	Analyst	Mark entity resolution as candidate, probable, confirmed, disputed or unresolved.	Entity	ENT-01,ASM-01	Graph/search visibly carries status.
F-ENT-006	External entity enrichment	P1	Analyst	Request authorised enrichment from external registries/screening services.	Entity,External Reference	SRC-01,PRI-01	Imported enrichment retains provider/source/time/licence metadata.

CAP-05 — Relationship / Ownership / Asset

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-REL-001	Relationship records	P0	Analyst	Create first-class typed relationship with evidence, dates, confidence and status.	Relationship	REL-01	Graph edge can be traced to relationship record and evidence.
F-REL-002	Ownership interest	P0	Analyst	Record direct/indirect ownership, percentage, dates and evidence.	OwnershipInterest	REL-01,AST-01	Unknown percentage is permitted; not coerced to zero.
F-REL-003	Control assertion	P0	Analyst	Represent legal owner, beneficial owner, controller, user and associate separately.	ControlAssertion	AST-01	UI never collapses ownership/control/use into one field.
F-AST-001	Asset registry	P0	Analyst	Create property, vehicle, shares, vessel, aircraft, crypto and other asset records.	Asset	AST-01	Asset attribution type and evidence are required for material claims.
F-AST-002	Valuation records	P1	Analyst	Record estimated/reported valuations with date, currency and source.	AssetValuation	AST-01	Valuation method and uncertainty are visible.
F-REL-004	Beneficial ownership chain	P1	Analyst	Traverse multi-level ownership/control chains and calculate indicative indirect interest when inputs are known.	OwnershipInterest,Relationship	REL-01,TEC-02	Calculated values display assumptions and source chain.

CAP-06 — Timeline & Events

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-TIM-001	Event records	P0	Analyst	Record appointments, incorporation, tender awards, purchases, transfers, court events and other time-bounded events.	Event	REL-01	Event date precision and evidence are explicit.
F-TIM-002	Visual timeline	P0	Analyst	View case/entity/asset events chronologically with filtering.	Event	AUD-01	Timeline items link back to canonical event and evidence.
F-TIM-003	Temporal relationship view	P1	Analyst	Show when ownership/directorship/control existed.	Relationship, Event	REL-01	Historical and current relationships are distinguishable.

CAP-07 — Value-Flow Analysis

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-VAL-001	ValueFlow record	P0	Analyst	Record origin, destination, value/range, currency, date, mechanism, evidence and class.	ValueFlow	VAL-01	flow_class is mandatory: DIRECT, DOCUMENTED, RECONSTRUCTED, or HYPOTHETICAL.
F-VAL-002	Multi-leg value flow	P0	Analyst	Build chains of contracts, payments, loans, transfers, purchases and conversions.	ValueFlow, ValueFlowLeg	VAL-01	Each leg retains separate evidence/confidence.
F-VAL-003	Flow visualisation	P0	Analyst	Visualise value movement without erasing flow class or uncertainty.	ValueFlow	VAL-01	Line style/label communicates class in UI and export.
F-VAL-004	Unknown / range values	P0	Analyst	Represent unknown, minimum, maximum and approximate values.	ValueFlow	VAL-01	System does not invent numeric zero for unknown value.
F-VAL-005	Flow reconciliation	P1	Analyst	Compare incoming/outgoing known values and identify gaps without implying illegality.	ValueFlow	TEC-02	Computed gap is labelled analytical calculation with inputs/version.

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-VAL-006	Currency normalisation	P1	Analyst	View original currency and optionally normalised value using dated FX reference.	ValueFlow	TEC-02	Original value/currency always preserved.

CAP-08 — Typology & Indicator Analysis

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-TYP-001	Typology catalogue browser	P0	Analyst	Browse CS-AML typologies, mechanisms, observables, indicators and false positives.	Typology	TYP-01	Catalogue entry version is visible.
F-TYP-002	Indicator capture	P0	Analyst	Link observed indicators/counter-indicators to entities, events, flows and evidence.	Indicator	TYP-01	Indicator cannot exist as unattributed free-floating accusation in final assessment.
F-TYP-003	Typology match worksheet	P0	Analyst	Compare observed case pattern to typology with consistency level and alternatives.	TypologyMatch	TYP-01,HYP-01	Single weak indicator cannot auto-produce strong match.
F-TYP-004	Custom/local typologies	P1	Admin/Analyst	Create versioned local typologies without modifying official catalogue history.	Typology	AUD-01	Custom typology has owner, version and source lineage.
F-TYP-005	Rule-assisted indicator suggestions	P2	Analyst	Suggest potential indicators from graph/data while requiring confirmation.	IndicatorCandidates	TEC-01,TEC-02	Suggestions are labelled machine/rule-generated and cannot auto-escalate case.

CAP-09 — Hypothesis & Assessment

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-HYP-001	Hypothesis workspace	P0	Analyst	Maintain multiple hypotheses including legitimate explanations.	Hypothesis	HYP-01	At least one alternative hypothesis is supported for high-impact adverse assessment unless rationale documented.

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-HYP-002	Evidence support / contradiction matrix	P0	Analyst	Link evidence and indicators as supporting, contradicting, neutral or unknown for each hypothesis.	Hypothesis,EvidenceItem,Indicator	HYP-01,HYP-02	Matrix records analyst and rationale.
F-HYP-003	Intelligence gaps	P0	Analyst	Record unknowns that could materially change assessment.	IntelligenceGap	GAP-01	Final product surfaces material unresolved gaps.
F-ASM-001	Assessment record	P0	Analyst	Write judgement, confidence, basis, alternatives, assumptions and gaps.	Assessment	ASM-01	Assessment cannot be final without confidence and evidence-linked basis.
F-ASM-002	Confidence scale	P0	Analyst	Apply controlled confidence language with rationale.	Assessment	ASM-01	Confidence changes are versioned.
F-ASM-003	Disconfirming search record	P0	Reviewer	Document what was done to search for evidence that weakens adverse findings.	Review,Hypothesis	HYP-02,QUA-01	High-impact product cannot pass review if no disconfirmation record/rationale.

CAP-10 — Search / Graph / Analytics

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-SCH-001	Full-text search	P0	Analyst	Search case-authorised documents/evidence and extracted text.	EvidenceItem, DerivedArtifact	SEC-01	Results enforce object-level permissions.
F-SCH-002	Faceted entity/object search	P0	Analyst	Filter by entity type, identifier, jurisdiction, case, source, date and status.	Entity,Source,Case	SEC-01	Search does not reveal restricted object existence where policy forbids it.
F-GRF-001	Graph exploration	P0	Analyst	Explore entities, relationships, assets, events and flows visually.	Entity,Relationship,Asset,ValueFlow	REL-01,SEC-01	Every material edge can display provenance/confidence.
F-GRF-002	Path finding	P1	Analyst	Find paths between two entities under permission and relationship filters.	Relationship	TEC-02	Path result is labelled analytical output, not proof of association beyond underlying edges.

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-GRF-003	Network metrics	P2	Analyst	Run centrality, community and cluster analytics as assistive signals.	DerivedAnalytics	TEC-02	Algorithm/version/input scope are recorded and outputs cannot become facts automatically.
F-SCH-003	Saved searches / watch queries	P1	Analyst	Save recurring case-scoped queries and notify when new authorised data matches.	SavedQuery	SEC-01	Watch query scope inherits case permissions and can be disabled.

CAP-11 — Intelligence Products & Review

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-PRD-001	Intelligence product templates	P0	Analyst	Generate Financial Intelligence Note, Entity Profile, Asset Profile, Network Analysis, Referral Package and Case Report.	IntelligenceProduct	ASM-01,DIS-01	Product embeds version, classification, author/reviewer and evidence index.
F-REV-001	Peer review workflow	P0	Reviewer	Comment, request change, approve or reject high-impact products.	Review	QUA-01	Final approval is attributable and separate from author where required.
F-REV-002	Red-team review checklist	P1	Reviewer	Challenge identity, evidence, alternatives, confidence and potential harm.	Review	QUA-01,HYP-02	Checklist result is retained with product version.
F-PRD-002	Product versioning / corrections	P0	Analyst	Issue revised/corrected versions without deleting prior product history.	IntelligenceProduct	AUD-01	Recipients can distinguish superseded from current versions.
F-PRD-003	Evidence index generation	P0	Analyst	Generate evidence/source references supporting material findings.	IntelligenceProduct,EvidenceItem	SRC-01,EVD-02	Every key fact can be traced from product to evidence.

CAP-12 — Dissemination & Referral

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-DIS-001	Dissemination approval	P0	Case Owner	Approve external sharing with handling classification and intended recipient.	Dissemination	DIS-01	External export is blocked until required approval is present.
F-DIS-002	Secure export package	P0	Analyst	Export approved report and evidence index with minimisation/redaction.	Dissemination, IntelligenceProduct	DIS-01,PRI-01	Export records included objects and excludes unauthorised restricted content.
F-DIS-003	Referral package	P0	Analyst	Produce structured referral containing subjects, facts, evidence, timeline, flows, indicators, gaps and contact point.	IntelligenceProduct,Dissemination	DIS-01	Referral distinguishes known facts from analysis and unknowns.
F-DIS-004	Sharing log	P0	Case Owner	Record who received what version, when, purpose and restrictions.	Dissemination	DIS-01,AUD-01	Log is immutable to ordinary analysts.
F-DIS-005	Partner workspace / federation	P2	Partner	Share selected objects/products with controlled partner environment.	SharedObject	SEC-01,DIS-01	Sharing is explicit object-level allowlist; no implicit case replication.

CAP-13 — Administration & Governance

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-ADM-001	Controlled vocabulary administration	P0	Admin	Manage relationship types, asset types, statuses and classifications with versioning.	Vocabulary	AUD-01	Historical records preserve original term/version semantics.
F-ADM-002	Template administration	P1	Admin	Configure case, review and report templates.	Template	AUD-01	Template changes are versioned and do not rewrite historical products.
F-ADM-003	Retention policies	P0	Admin/Privacy	Configure retention/disposition by classification and object type.	RetentionPolicy	PRI-02	Policy execution can be previewed and audited.

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-ADM-004	Feature flags / environment policy	P1	Platform Admin	Enable high-risk capabilities (AI, external connectors, federation) per deployment.	SystemPolicy	SEC-01	Disabled capability cannot be bypassed through API.

CAP-14 — Security / Audit / Operations

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-SEC-001	OIDC authentication + MFA	P0	All users	Authenticate through organisation identity provider with MFA where required.	User,Session	SEC-01	Disabled user loses access; session policy enforced.
F-SEC-002	Role and object-level access control	P0	Security Admin	Restrict access by role, case membership, classification and need-to-know.	AccessPolicy	SEC-01,SEC-02	API and UI enforce the same authorisation decision.
F-SEC-003	Protected-source compartment	P0	Source Handler	Store protected source identity separately with restricted access.	ProtectedSource	SEC-02	Routine analysts can use source-derived evidence without automatically seeing identity.
F-AUD-001	Immutable audit trail	P0	Auditor	Record material create/update/delete/merge/review/export actions.	AuditEvent	AUD-01	Ordinary users cannot edit audit events.
F-OPS-001	Backup and restore	P0	Security Admin	Back up canonical DB, evidence and configuration; test restore.	BackupRecord	SEC-01	Restore test is documented at defined interval.
F-OPS-002	Operational monitoring	P1	Platform Admin	Monitor health, errors, jobs, capacity and security events without logging sensitive evidence content.	OperationalEvent	SEC-01	Observability redacts/avoids case-sensitive payloads.
F-OPS-003	Incident response support	P1	Security Admin	Record security incidents affecting sources, evidence, accounts or exports.	SecurityIncident	SEC-02	Incident workflow supports containment, notification and lessons learned.

CAP-15 — Integrations & Automation

Feature ID	Feature	Pri	Role	Requirement / user story	Data objects	Control refs	Acceptance criterion
F-INT-001	OpenAleph connector	P1	Analyst	Link/import investigative documents/entities from OpenAleph with provenance.	ExternalReference,Entity,Source	SRC-01	Imported objects retain remote IDs and source metadata; no silent merge.
F-INT-002	OpenSanctions screening connector	P1	Analyst	Screen entity against sanctions/PEP/debarment data and review candidate matches.	ScreeningResult	ENT-01,TEC-01	Screening result is a candidate signal, not canonical identity/fact until reviewed.
F-INT-003	FollowTheMoney mapping	P1	Data Steward	Map compatible entities/ownership/payment concepts to/from FollowTheMoney.	Entity,Relationship,ValueFlow	AUD-01	Import/export mapping is versioned and preserves unmapped fields.
F-INT-004	Flowintel interoperability	P2	Case Owner	Link task/case workflow metadata with Flowintel where used.	ExternalCaseReference	AUD-01	No implicit transfer of sensitive evidence without explicit configuration.
F-INT-005	GraphSense connector	P2	Analyst	Enrich public-chain wallet/address analysis using GraphSense.	ExternalReference,ValueFlow	TEC-01	Blockchain analytics output remains attributed to provider and timestamp.
F-INT-006	Graph projection connector	P1	Platform Admin	Project canonical relationship/value-flow data to Neo4j/Memgraph or equivalent.	DerivedProjection	REL-01	Projection is rebuildable and cannot become sole canonical source.
F-AI-001	AI summarisation / extraction assist	P1	Analyst	Use AI to summarise or propose entities/relationships from authorised content.	DerivedArtifact,Candidate	TEC-01	AI output is labelled generated, attributable to model/version, and review-required.
F-AI-002	AI analytical assistant	P2	Analyst	Ask questions across authorised case data with citations to canonical objects.	DerivedAnalyses	TEC-01,TEC-02	Assistant cannot create final adverse assessment or dissemination approval autonomously.

9. MVP Definition (Release 0.1)

The MVP SHALL demonstrate the complete analytical chain on a real or synthetic civil-society investigation without depending on advanced automation. Every P0 feature is in scope unless explicitly waived in a release decision.

Module	Minimum outcome
MVP-1 Case Workspace	Case register, charter, lifecycle, tasks, ownership, status, audit
MVP-2 Source & Evidence	Source register, preserved evidence, hashes, extracts, derivatives, reliability
MVP-3 Entity & Relationship	Canonical entities, aliases/identifiers, candidate match, merge/unmerge, first-class relationships
MVP-4 Assets / Events / Timeline	Asset attribution, event records, visual timeline
MVP-5 Follow-the-Value	ValueFlow builder with mandatory DIRECT/DOCUMENTED/RECONSTRUCTED/HYPOTHETICAL classification
MVP-6 Typology & Hypothesis	Catalogue browser, indicators, typology worksheet, competing hypotheses, gaps
MVP-7 Assessment & Review	Assessment, confidence, disconfirmation, peer review
MVP-8 Intelligence Product	Report templates, evidence index, versioning, secure export/referral
MVP-9 Search & Graph	Full-text/faceted search and provenance-aware graph view
MVP-10 Security & Audit	OIDC/MFA, object-level RBAC, protected-source compartment, audit, backup/restore

MVP completion test

A two-analyst team SHALL be able to open a case, ingest evidence, resolve entities, build an ownership/asset/value-flow model, test at least two hypotheses, map relevant typologies, produce a confidence-rated assessment, complete independent review, and export an approved intelligence product while preserving provenance and audit history.

10. Release Roadmap

Release	Scope	Exit objective
0.1 — Foundation MVP	P0 features	Complete end-to-end investigation and review workflow with strong provenance and security.
0.2 — Operational Intelligence	P1 features	OCR/extraction, enrichment connectors, advanced ownership, saved queries, red-team review, observability.
0.3 — Network Intelligence	Selected P2	Cross-case discovery, network metrics, GraphSense, federation/partner workspace, rule-assisted indicators.
0.4 — Controlled AI	Selected P1/P2 AI	Citation-grounded AI assistance, controlled entity suggestions, model governance, validation metrics.

Release	Scope	Exit objective
1.0 — Assured Platform	All required controls + independent QA	Production-ready product with conformance evidence, security assurance, documented operations and upgrade path.

11. Build vs Integrate Strategy

Disposition	Capabilities	Rationale
Build as CS-AML core	Case charter/gates; Evidence→Fact chain; hypothesis matrix; confidence; intelligence gaps; value-flow classification; typology assessment; review/dissemination; provenance-preserving UX	These functions embody the unique methodology and control model.
Integrate / reuse	OCR/document intelligence; sanctions/PEP data; generic graph DB; blockchain analytics; identity provider; search engine; object storage; observability	Mature components already exist; wrapping them preserves focus and reduces maintenance risk.
Adapt / map	FollowTheMoney ontology; OpenAleph investigative datasets; Flowintel workflow concepts	Useful models/workflows can be mapped without making them normative dependencies.

Tool / ecosystem	Reference value	Recommended role	Phase
OpenAleph	Investigative document/data platform; reference for collections, search, OCR/entity extraction, investigation workspaces.	Connector or adjacent investigative data platform	P1
FollowTheMoney	Investigative entity/relationship ontology used in anti-corruption and investigative journalism ecosystems.	Interoperability / ontology mapping	P1
OpenSanctions / yente	Sanctions, PEP and related entity matching/screening.	External screening service	P1
Flowintel	Open-source case/task workflow platform.	Workflow interoperability / UX reference	P2
GraphSense	Open-source cryptoasset analytics.	Crypto analytics connector	P2
Neo4j / Memgraph	Graph database / analytics.	Derived graph projection	P1/P2
OpenSearch / PostgreSQL FTS	Search engine.	Derived search index	P0/P1
Keycloak or equivalent	OIDC IAM, MFA, RBAC support.	Identity provider	P0
S3-compatible object store	Evidence object storage.	Evidence storage infrastructure	P0
OpenTelemetry + Prometheus/Grafana	Telemetry and operational monitoring.	Observability	P1

12. Non-Functional Requirements

ID	Domain	Requirement
NFR-SEC-01	Security	Least privilege, MFA, compartmentalisation, encryption in transit, secure secrets, patching, backup, audit.
NFR-PRI-01	Privacy	Data minimisation, purpose limitation, retention, deletion/disposition, access justification, export minimisation.
NFR-AUD-01	Auditability	Material analytical and dissemination changes are attributable, timestamped and reviewable.
NFR-INT-01	Integrity	Canonical objects and original evidence cannot be silently overwritten; critical evidence supports hash verification.
NFR-EXP-01	Explainability	Automated score/match/path/suggestion used materially exposes basis, input scope and model/rule version.
NFR-PER-01	Performance	Common case, entity, search and graph views SHOULD return interactively for expected deployment scale; long analytics MAY use async jobs.
NFR-AVL-01	Availability	System SHOULD tolerate component restart and provide tested backup/restore; deployment defines RPO/RTO.
NFR-INTOP-01	Interoperability	Stable IDs, versioned APIs, import/export metadata and mapping records SHALL support external integration.
NFR-UX-01	Usability	UX SHALL make fact/inference/confidence/provenance visible without requiring analyst to inspect raw DB fields.
NFR-ACC-01	Accessibility	Core workflows SHOULD meet recognised web accessibility good practice and support keyboard/screen-reader use.
NFR-MNT-01	Maintainability	Schema, APIs, typologies, vocabularies and templates are versioned; migrations are testable and reversible where feasible.
NFR-OBS-01	Observability	Operational telemetry SHALL avoid leaking protected evidence or source content into logs.

13. Permission and Information-Handling Model

At minimum, authorisation SHALL consider role, case membership, object classification, need-to-know, protected-source compartment, and dissemination restriction. Case membership alone SHALL NOT automatically reveal protected-source identity.

Classification	Handling expectation
Public / Open	Suitable for broad internal visibility or public source references; still subject to case purpose and privacy rules.

Classification	Handling expectation
Internal	Routine investigation content limited to authorised organisation users.
Restricted	Sensitive personal, legal-risk, unpublished, or partner-controlled information.
Protected Source	Source identity/contact and highly sensitive source-handling data; separate compartment and minimal access.
Legal / Need-to-Know	Material with legal privilege, safety, embargo, or highly constrained access conditions.

14. Product Data and API Requirements

Requirement	Rule
Stable object IDs	All canonical objects SHALL expose stable identifiers suitable for references, audit and integration.
Version metadata	Material objects SHALL expose current version and modification metadata; previous versions remain reconstructable where required.
Provenance links	APIs SHALL allow traversal from assessment to hypothesis/indicator/fact/evidence/source.
Permission enforcement	API SHALL enforce the same authorisation rules as UI; UI hiding is not a security control.
Derived output labelling	Search hits, graph paths, AI suggestions and computed metrics SHALL expose derived status and source inputs.
Export manifests	Exports SHOULD contain object/version manifest and handling metadata.
Connector lineage	Imported records SHALL retain provider, external ID, retrieval time, connector version and mapping status.

15. Required User Experience Surfaces

Surface	Minimum UX purpose
Dashboard	Assigned cases, pending reviews, high-priority tasks, recent activity, connector/processing warnings.
Case Workspace	Charter, scope, tasks, linked entities/evidence, timeline, hypotheses, assessments, reviews, products.
Evidence Viewer	Original/derivative distinction, page/text view, extract creation, hash/provenance, linked claims/facts.
Entity Profile	Aliases, identifiers, source assertions, related cases, relationships, assets, events, screening/enrichment.
Entity Resolution Workbench	Side-by-side candidate comparison, matching/conflicting attributes, evidence, merge/unmerge decision.
Graph Explorer	Provenance-aware nodes/edges, filters, saved views, path query, evidence drill-down.
Timeline	Events, relationship validity, asset changes, flow dates and evidence.
Value Flow Builder	Leg creation, class distinction, amount/range/currency, evidence, flow visualisation.
Typology Worksheet	Catalogue guidance, indicators, counter-indicators, consistency assessment and references.

Surface	Minimum UX purpose
Hypothesis Matrix	Competing hypotheses, supporting/contradicting evidence, gaps, discriminating questions.
Assessment Editor	Judgement, confidence, basis, alternatives, assumptions, gaps, reviewer status.
Product / Review Workspace	Template, evidence index, comments, red-team checklist, approval, versioning.
Admin / Assurance	Users/roles, vocabularies, connectors, policy, retention, audit, control evidence and system health.

16. Suggested Engineering Epics

Epic	Name	Scope
EPIC-01	Foundation platform	Project skeleton, auth, tenancy/deployment policy, canonical IDs, audit framework.
EPIC-02	Case & workflow	Case charter, tasks, gates, ownership, lifecycle.
EPIC-03	Evidence & provenance	Source/evidence registry, file store, hash, extracts, derivatives.
EPIC-04	Entity model & resolution	Entity registry, identifiers, match candidates, merge/unmerge.
EPIC-05	Relationships/assets/events	First-class graph objects, ownership/control, assets, event/timeline.
EPIC-06	Value-flow	Flow/legs, visual builder, classifications, calculations.
EPIC-07	Typology & hypothesis	Catalogue, indicators, hypothesis matrix, gaps.
EPIC-08	Assessment & products	Confidence, intelligence product, evidence index, versioning.
EPIC-09	Review & dissemination	Peer review, red-team, approval, secure export, referral, sharing log.
EPIC-10	Search & graph	FTS, faceted search, graph projection/explorer, saved views.
EPIC-11	Security & operations	RBAC, protected source, backup/restore, monitoring, incident support.
EPIC-12	Integrations	OpenAleph, OpenSanctions, FTM, GraphSense, external graph/search.
EPIC-13	AI assist	Controlled extraction, summarisation, grounded assistant, model audit.

17. Acceptance and Test Strategy

Test domain	Minimum acceptance approach
Functional acceptance	Feature-specific acceptance criteria in Section 8 SHALL be converted to executable/manual test cases.
Methodology conformance	Test that users cannot silently skip mandatory separation between evidence, fact, hypothesis and assessment.
Permission testing	Include positive and negative tests for case/object/source compartment access.

Test domain	Minimum acceptance approach
Provenance testing	Randomly sample material findings and trace backward to evidence/source.
Entity-resolution testing	Test false-positive merges, conflicting identifiers, reversal, and historical integrity.
Value-flow testing	Verify all four flow classes remain distinct in DB/API/UI/export.
Review testing	High-impact product cannot be finalised without required independent review.
Audit testing	Critical action creates audit event and ordinary users cannot edit/delete it.
Connector testing	External enrichment failure or stale data cannot corrupt canonical objects.
AI testing	AI output is visibly derived, grounded where applicable, and blocked from autonomous adverse decision or dissemination.
Backup/restore testing	Restore canonical DB and evidence into isolated environment and verify object/evidence integrity.

18. Product Metrics

Metric	Interpretation
Provenance coverage	% material facts/relationships/value flows with direct evidence/source links.
Entity correction rate	Merge/unmerge correction rate and false-match review outcome.
Review coverage	% high-impact products independently reviewed before dissemination.
Gap visibility	% material assessments with explicit intelligence gaps.
Alternative-hypothesis coverage	% high-impact adverse assessments with documented alternatives/disconfirmation.
Time-to-triage	Median lead intake to open/reject/monitor decision.
Time-to-review	Median draft product to completed peer review.
Referral usefulness	Recipient feedback or follow-up rate where legitimately measurable.
Security/privacy incidents	Source exposure, unauthorised access, accidental export or retention exceptions.
Analyst rework	Corrections caused by missing provenance, identity error, unsupported relationship or ambiguous flow classification.

Metric warning

Node count, scraped-record count, number of alerts, AI-generated suggestions, or number of cases are not sufficient measures of effectiveness.

19. Framework-to-Product Traceability

Framework requirement	Feature implementation	Data objects	Controls
Evidence before allegation	F-EVD-*; F-HYP-*; F-ASM-*	Source/Evidence/Fact/Hypothesis/Assessment	SRC/EVD/HYP/ASM
Case is context; reusable truth objects	F-CASE-*; F-ENT-*; F-EVD-*	Case/Entity/Evidence	CAS/ENT/EVD
Follow the value	F-VAL-*	ValueFlow/ValueFlowLeg	VAL-01

Framework requirement	Feature implementation	Data objects	Controls
Typology caution	F-TYP-*	Typology/Indicator/ TypologyMatch	TYP-01
Entity resolution reversibility	F-ENT-003/004/005	EntityMatchCandidate/ MergeDecision	ENT-01
Alternative explanations	F-HYP-001/002/003	Hypothesis/IntelligenceGap	HYP-01/HYP-02/GAP-01
Confidence and assessment	F-ASM-001/002	Assessment	ASM-01
Peer review	F-REV-*	Review	QUA-01
Controlled dissemination	F-DIS-*	Dissemination/ IntelligenceProduct	DIS-01
Protected sources	F-SEC-003	ProtectedSource	SEC-02
AI is assistive	F-AI-*; F-DOC-004/006	DerivedArtifact/Candidate	TEC-01/TEC-02

20. Open Product Decisions for v0.1

ID	Decision	Recommended v0.1 stance
PD-01	Single-organisation first vs multi-tenant from day one	Recommendation: single-organisation / strong case compartment first; design schema for later multi-tenancy.
PD-02	OpenAleph as dependency vs connector	Recommendation: connector/adjacent service, not hard dependency, until workflow is validated.
PD-03	FollowTheMoney adoption depth	Recommendation: interoperability mapping and selective reuse; retain CS-AML-specific Evidence/Hypothesis/Assessment model.
PD-04	Graph database in MVP	Recommendation: optional projection. MVP can use relational canonical model plus graph view; introduce dedicated graph DB when scale/queries justify it.
PD-05	AI in MVP	Recommendation: not required for MVP completion. Introduce after evidence/provenance and review workflow are stable.
PD-06	Public-source connectors	Recommendation: begin with manual import + a small number of legally reviewed connectors; avoid uncontrolled scraping framework in v0.1.
PD-07	Mobile support	Recommendation: responsive read/review/task capability first; sensitive collection and evidence handling optimised for desktop.
PD-08	Federation	Recommendation: defer to Phase 3; first establish object-level sharing and export manifests.

Annex A — Canonical User Story Format

Each backlog item SHOULD use the following fields:

- Feature ID / Epic ID
- As a <role>
- I want <capability>
- So that <investigative/control outcome>
- Preconditions
- Primary flow
- Alternative/error flows
- Data objects touched
- Control references
- Security/privacy considerations
- Acceptance criteria
- Audit events
- Telemetry (non-sensitive)
- Priority / release
- Dependencies

Annex B — Definition of Done for P0 Features

- Functional acceptance criteria passed.
- Authorisation tested for permitted and denied users.
- Audit events verified for material actions.
- Provenance / object references preserved.
- No silent conversion of derived or inferred data into canonical fact.
- Error states fail safely and do not corrupt originals/canonical objects.
- API and UI behaviour are consistent.
- Retention/classification behaviour tested where applicable.
- Backup/migration impact documented for schema changes.
- User-facing wording distinguishes fact, inference, confidence and uncertainty.
- Operational documentation and release notes updated.

Annex C — Reference Product Architecture (Non-Normative)

A pragmatic first implementation MAY use the following profile:

Layer	Reference option
Web application / API	Django + REST/GraphQL equivalent
Canonical relational store	PostgreSQL + PostGIS where geographic data is needed
Evidence object store	S3-compatible storage or encrypted filesystem
Async jobs	Celery/RQ equivalent + Redis-compatible broker
IAM	Keycloak or other OIDC provider
Search	PostgreSQL FTS initially; OpenSearch when scale/features justify it
Graph	Relational projection initially; Neo4j/Memgraph optional derived projection
OCR/document processing	OpenAleph or dedicated processing service

Layer	Reference option
Sanctions/PEP screening	OpenSanctions/yente connector
Crypto analytics	GraphSense connector
Observability	OpenTelemetry + Prometheus/Grafana equivalent
Reverse proxy	Nginx/Caddy equivalent

Annex D — Release Planning Summary

Priority	Feature count	Planning meaning
P0 / MVP	54	Required to prove end-to-end methodology and controls
P1 / Phase 2	26	Operational depth, enrichment, advanced review and analytics
P2 / Phase 3	7	Network intelligence, federation, advanced AI/analytics